FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

3

CASE-BASED REASONING (CBR) SYSTEM DEVELOPMENT

Student Name	Anonuevo, Kassandra Margarette
Section:	BSCSSE – AN43
Professor:	Tan, Crisola
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I. BACKGROUND INFORMATION

Case-Based Reasoning (CBR) is an intelligent problem-solving approach used in **Knowledge Representation and Reasoning (KRR)**. Instead of relying solely on predefined rules, CBR systems solve new problems by referring to past experiences, also known as cases. These cases contain previous problems and their successful solutions. When a new situation arises, the system identifies the most similar past case, adapts the old solution to fit the new context, and applies it. Afterward, the new case is saved, allowing the system to learn and continuously improve.

CBR is widely used in various fields, such as disaster response, legal assistance, and intelligent tutoring, because it mimics how humans naturally make decisions. For example, emergency responders rely on past typhoon data to plan evacuations; lawyers examine previous legal cases to build arguments; and tutoring systems look at prior student mistakes to provide customized guidance. In all these examples, past knowledge becomes the foundation for solving future problems more effectively.

This activity allows students to experience how CBR works by developing a simple, functional CBR program based on real-world scenarios. Through pair programming, students will learn how to represent cases, implement similarity assessment, adapt solutions, and store new learning experiences—reflecting the complete CBR cycle. This hands-on approach strengthens understanding of how intelligent systems use experiential knowledge to support reasoning, decision-making, and learning.

II. TECHNICAL ASSESSMENT

CASE-BASED REASONING (CBR) SYSTEM DEVELOPMENT

Objective

At the end of this laboratory activity, students are expected to:

1. **Understand and apply the Case-Based Reasoning (CBR) cycle**, including retrieval, reuse, revision, and retention.
2. **Represent cases effectively** using appropriate data structures such as dictionaries, arrays, JSON objects, or classes.
3. **Implement a similarity assessment method** to identify relevant past cases based on defined attributes.
4. **Adapt retrieved solutions** to meet the needs of a new problem scenario based on the chosen case study.
5. **Demonstrate learning in CBR** by saving the newly solved case into the system's case base.
6. **Develop collaboration skills** through pair programming and joint problem analysis.
7. **Apply knowledge representation and reasoning concepts** to real-world problems in disaster response, legal reasoning, or intelligent tutoring.

Materials Needed

1. Hardware
    - A laptop or desktop computer
    - Stable internet connection (for research, documentation, submission)
2. Software

A programming environment:

- Python (recommended: IDLE, VS Code, Jupyter Notebook, PyCharm)
- Any text editor (Notepad++, Sublime, VS Code) if needed

3. File Formats & Tools

Word processor for documentation:

- MS Word, Google Docs, or LibreOffice
- Screenshot tool (e.g., Snipping Tool, Lightshot, Print Screen)

4. Reference Materials

Lecture notes on:

- Knowledge Representation & Reasoning
- Case-Based Reasoning cycle
- Similarity measures

Provided case studies (Disaster DSS, Legal DSS, Intelligent Tutoring System)

5. Pair Collaboration Tools (optional but useful)
    - Google Drive for shared documentation
    - GitHub/GitLab for code collaboration

General Instructions

1. This **Technical Assessment (TA3)** must be completed by Pair.
2. Please ensure that you are grouped as pairs in the People tab.
3. Select only one case study from the list below.
4. Include clear comments in your code to describe its logic.
5. Each program should display the correct input, process, and output.
6. Ensure that your script displays clear and readable outputs.
7. Compile all Python codes and corresponding outputs into one PDF file.
8. Save it as **TA3_lastname1_lastname2.pdf**
   (Example: TA2_Patilano_Loverez.pdf)
9. Submit the **python file** and **PDF** file.

📌 Step-by-Step Instructions

1. Form a pair

Choose a partner. Each pair will work collaboratively on research, coding, debugging, and presenting your results.

2. Choose ONE (1) case study

Select only one case study from the list below. Your program must be based on the scenario and logic of the chosen system:

Case Study Options:

1. *Disaster Response Decision Support System (DSS)*
2. *AI-Based Legal Decision Support System*
3. *Intelligent Tutoring System (ITS)*

3. Define your case base

Create at least **3–5 past cases** stored as dictionaries, JSON objects, arrays, or any structure.

Example structure:

```
{
  "problem": {...},
  "solution": {...}
}
```

4. Implement similarity assessment

Your program must include a way to measure similarity between the new case and past cases. Examples:

- Attribute matching
- Weighted scoring
- Simple rule-based comparison

Output should show: (sample)

✓ Similarity score or ranking

✓ The most relevant past case retrieved

5. Adapt the solution

Modify the retrieved solution so it fits the new problem scenario.
This part should reflect the logic of your chosen case study:

Examples:

- **DSS:** Adjust evacuation plans based on population size
- **Legal DSS:** Modify arguments due to new digital evidence
- **ITS:** Give personalized feedback based on student errors

6. Demonstrate learning (retain stage)

After generating a final solution, store the new case into your case base.

Your program must show:

✓ "New case saved" or similar confirmation

✓ Updated case list

7. Display clear program outputs

Your output must include:

- New problem description
- Retrieved similar case

- Adapted final solution
- Updated case base

8. Submission Requirements

Submit the following:

A. Program File

- Python

B. Short Documentation (1–2 pages)

Include:

- Title of chosen case study
- Description of your case base
- Explanation of similarity method
- Adaptation logic
- Screenshots of program execution

C. Pair Reflection (5–7 sentences)

Discuss:

- What each partner contributed
- What challenges you encountered
- What you learned about CBR

Case Studies:

Case Study 1: Disaster Response Decision Support System

A CBR-based DSS helps local government units respond to natural disasters.

- Past cases include typhoons, floods, and earthquakes.
- New event: *Typhoon with expected flooding in urban areas.*
- System retrieves similar events: *Typhoon Ursula (2019)* and *Typhoon Odette (2021)*.
- It adapts recommended responses such as evacuation strategy, resource allocation, and shelter capacity.
- After execution, the system stores new outcomes for improved future reasoning.

Learning Outcome:

CBR helps governments respond faster by using *experiential knowledge* rather than rebuilding decisions from scratch.

Case Study 2: AI-Based Legal Decision Support System

A legal CBR tool assists lawyers by retrieving previous court cases similar to the current legal issue.

Scenario:

Client is charged with "theft involving electronic transfer." System retrieves past cases involving:

- online theft
- digital fraud
- unauthorized bank transfers

Adaptation:

The reasoning engine adjusts argument patterns to match the new technical context.

Learning:

After final verdict, the case is stored—enhancing the system's knowledge for future cybercrime-related

offenses.

Case Study 3: Intelligent Tutoring System
A CBR system identifies the learning struggles of students. Student submits a math solution with repeated errors. System retrieves past students with similar error patterns (e.g., misusing formulas). It adapts feedback and personalized lessons based on successful interventions. **Learning:** System improves by retaining new student profiles and outcomes.

Your Code - Laboratory Tasks ⇩

```
# TA3 - Mathematics Intelligent Tutoring System
# Case-Based Reasoning Program
#Kassandra A. | 20231000026 | AN43

# ----------------------------------------
# CASE BASE
# ----------------------------------------

case_base = [
    {
        "topic": "Quadratic Equations",
        "error": "Misusing the quadratic formula",
        "solution": "Used b instead of -b in the formula.",
        "intervention": "Review the quadratic formula and identify a, b, and c.",
        "outcome": "Improved"
    },
    {
        "topic": "Quadratic Equations",
        "error": "Wrong substitution",
        "solution": "Substituted numbers into the wrong parts of the formula.",
        "intervention": "Organize a, b, and c before substituting.",
        "outcome": "Improved"
    },
    {
        "topic": "Linear Equations",
        "error": "Arithmetic error",
        "solution": "Made an error while adding and subtracting numbers.",
        "intervention": "Solve one operation at a time and check every step.",
        "outcome": "Improved"
    },
    {
```

```
        "topic": "Factoring",
        "error": "Incorrect factors",
        "solution": "Selected factors that did not match the equation.",
        "intervention": "Practice identifying factor pairs.",
        "outcome": "Needs more practice"
    }
]


# ----------------------------------------
# SELECT MATH TOPIC
# ----------------------------------------

print("MATHEMATICS INTELLIGENT TUTORING SYSTEM")
print("----------------------------------------")

print("\nChoose a math topic:")
print("1. Quadratic Equations")
print("2. Linear Equations")
print("3. Factoring")

topic_choice = input("Enter your choice: ")

if topic_choice == "1":
    topic = "Quadratic Equations"
elif topic_choice == "2":
    topic = "Linear Equations"
elif topic_choice == "3":
    topic = "Factoring"
else:
    topic = "Quadratic Equations"
    print("Invalid choice. Quadratic Equations selected.")


# ----------------------------------------
# SELECT ERROR PATTERN
# ----------------------------------------

print("\nChoose the repeated error:")
print("1. Misusing the quadratic formula")
print("2. Wrong substitution")
print("3. Arithmetic error")
print("4. Incorrect factors")
```

```
error_choice = input("Enter your choice: ")

if error_choice == "1":
    error = "Misusing the quadratic formula"
elif error_choice == "2":
    error = "Wrong substitution"
elif error_choice == "3":
    error = "Arithmetic error"
elif error_choice == "4":
    error = "Incorrect factors"
else:
    error = "Misusing the quadratic formula"
    print("Invalid choice. Misusing the quadratic formula selected.")


solution = input("\nEnter the student's submitted math solution: ")


# ----------------------------------------
# SIMILARITY ASSESSMENT
# ----------------------------------------

def calculate_similarity(new_case, old_case):
    score = 0

    if new_case["topic"] == old_case["topic"]:
        score += 1

    if new_case["error"] == old_case["error"]:
        score += 1

    return score


new_case = {
    "topic": topic,
    "error": error,
    "solution": solution
}

results = []
```

```
for old_case in case_base:
    score = calculate_similarity(new_case, old_case)
    results.append((score, old_case))

results.sort(reverse=True, key=lambda item: item[0])

highest_score, retrieved_case = results[0]


# ----------------------------------------
# ADAPTATION STAGE
# ----------------------------------------

if error == "Misusing the quadratic formula":
    feedback = (
        "Review the quadratic formula. Identify a, b, and c, "
        "then substitute each value carefully. Pay attention to negative signs."
    )

elif error == "Wrong substitution":
    feedback = (
        "Write the values of a, b, and c before substituting "
        "them into the formula."
    )

elif error == "Arithmetic error":
    feedback = (
        "Solve one operation at a time and check your addition "
        "and subtraction after every step."
    )

else:
    feedback = (
        "List possible factor pairs and check them using multiplication "
        "and addition."
    )


# ----------------------------------------
# OUTCOME CHOICE
# ----------------------------------------

print("\nStudent outcome:")
```

```
print("1. Improved")
print("2. Needs more practice")

outcome_choice = input("Enter your choice: ")

if outcome_choice == "1":
    outcome = "Improved"
else:
    outcome = "Needs more practice"


# ----------------------------------------
# RETENTION STAGE
# ----------------------------------------

new_case["feedback"] = feedback
new_case["outcome"] = outcome

case_base.append(new_case)


# ----------------------------------------
# OUTPUT
# ----------------------------------------

print("\nCASE-BASED REASONING RESULTS")
print("----------------------------")

print("\nNEW STUDENT PROBLEM")
print(f"Math Topic: {topic}")
print(f"Repeated Error: {error}")
print(f"Submitted Solution: {solution}")

print("\nRETRIEVED SIMILAR CASE")
print(f"Previous Topic: {retrieved_case['topic']}")
print(f"Previous Error: {retrieved_case['error']}")
print(f"Previous Solution: {retrieved_case['solution']}")
print(f"Previous Intervention: {retrieved_case['intervention']}")
print(f"Similarity Score: {highest_score}/2")

print("\nADAPTED PERSONALIZED FEEDBACK")
print(feedback)
```

```
print("\nSTUDENT OUTCOME")
print(outcome)

print("\nRETENTION")
print("New student profile and outcome saved.")
print(f"Updated case base: {len(case_base)} cases")

print("\nCBR CYCLE COMPLETED")
print("Retrieval -> Reuse -> Revision -> Retention")
```

SCREENSHOTS:

```
MATHEMATICS INTELLIGENT TUTORING SYSTEM
----------------------------------------

Choose a math topic:
1. Quadratic Equations
2. Linear Equations
3. Factoring
Enter your choice: 1

Choose the repeated error:
1. Misusing the quadratic formula
2. Wrong substitution
3. Arithmetic error
4. Incorrect factors
Enter your choice: 1

Enter the student's submitted math solution: I used b instead of -b in the formula.

Student outcome:
1. Improved
2. Needs more practice
Enter your choice: 1

CASE-BASED REASONING RESULTS
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NEW STUDENT PROBLEM
Math Topic: Quadratic Equations
Repeated Error: Misusing the quadratic formula
Submitted Solution: I used b instead of -b in the formula.

RETRIEVED SIMILAR CASE
Previous Topic: Quadratic Equations
Previous Error: Misusing the quadratic formula
Previous Solution: Used b instead of -b in the formula.
Previous Intervention: Review the quadratic formula and identify a, b, and c.
Similarity Score: 2/2
```

```
ADAPTED PERSONALIZED FEEDBACK
Review the quadratic formula. Identify a, b, and c, then substitute each value carefully. Pay attention to negative signs.

STUDENT OUTCOME
Improved

RETENTION
New student profile and outcome saved.
Updated case base: 5 cases

CBR CYCLE COMPLETED
Retrieval -> Reuse -> Revision -> Retention
```